MOJTABA ASHOUR

Ph.D. Candidate

Cornell University
4231 Martha Van Rensselaer Hall, Ithaca, NY 14850
(607) 229-2504 | ma2236@cornell.edu | www.dail.human.cornell.edu | Google Scholar

SUMMARY

Design researcher investigating how environmental design influences cognition, affect, and behavior, with the goal of translating knowledge of human-environment relationships into evidence-informed design. My current research focuses on wayfinding and spatial cognition in complex healthcare environments and extends across physical, virtual, and augmented settings using experimental, qualitative, and mixed-methods approaches. Author of 12 peer-reviewed journal publications and experience leading studies within NSF-funded research and contributing to NIH-funded research. My academic experience further encompasses research leadership and mentorship of graduate and undergraduate researchers, as well as teaching across design studio, seminar, and lecture formats. Three years of professional interior design practice complement my academic work and inform my approach to bridging research and design practice.

EDUCATION

2023-pres. Ph.D. in Human Behavior and Design, **Cornell University,** Ithaca, USA.
Department of Human Centered Design, College of Human Ecology
Saleh Kalantari (Chair); Adam Anderson (Cognitive Science); Nancy M. Wells (Ecology of Human Development)

2019-2021 Master of Science in Interior Design, **University of Science Malaysia,** Penang, Malaysia.
School of Housing, Building and Planning

2016-2019 Bachelor of Science in Housing, Building and Planning, **University of Science Malaysia,** Penang, Malaysia
School of Housing, Building and Planning

2011-2014 Diploma in Interior Design, **UOW Malaysia KDU Penang University College,** Penang, Malaysia
School of Built Environment

PROFESSIONAL EXPERIENCE

2026-pres. **Lab Manager**
Design and Augmented Intelligence Lab (DAIL), **Cornell University,** Ithaca, USA.

- Coordinate graduate and undergraduate researchers across multiple concurrent projects, including task assignment, scheduling, progress tracking, and deliverable oversight.
- Lead onboarding and training of new team members, developing workflows and providing instruction in research protocols, data management, literature review, qualitative and quantitative methods.
- Mentor graduate and undergraduate researchers through one-on-one and group meetings, providing methodological guidance, research supervision, and academic and professional development.
- Selected research mentees: Jialin Wu (graduate), Ariana Dahi (undergraduate), Yifei Gao (graduate), and Wenqian Niu (graduate); mentorship resulting in co-authored manuscripts and conference presentations.

2023-pres. **Research Assistant**
Design and Augmented Intelligence Lab (DAIL), **Cornell University,** Ithaca, USA.

- Developed research protocols and analysis plans and coordinated complex human-subjects research across experimental, observational, and evidence-synthesis studies.
- Conducted mixed-methods empirical research on human behavior, spatial cognition, and environmental experience across augmented reality, fully immersive virtual reality, and real-world environments.
- Led multiple studies within an NSF-funded research project on healthcare wayfinding, overseeing study design, development of standard operating procedures, participant recruitment and data collection, quantitative and qualitative analysis, and peer-reviewed publications and conference presentations.
- Contributed to the NIH R01-funded SERVE VR randomized controlled trial on social virtual environments for older adults, supporting experimental design, development of manuals of operating procedures, data analysis, and manuscript preparation.
- Led and contributed to multiple PRISMA-registered systematic reviews and meta-analyses on human behavior and design across immersive and real-world environments.

2020-2021 **Research Assistant**
School of Housing, Building and Planning, **University of Science Malaysia**, Penang, Malaysia.

- Conducted research on barriers to adopting sustainable design practices in Malaysia as part of Grant No. 304/PPBGN/6315356, in collaboration with national and international researchers.
- Contributed to peer-reviewed publications through scholarly writing, manuscript development and editing, data visualization, and journal submission, resulting in multiple publications on sustainable interior architecture, green design adoption, and decision-making methods.

2019-2020 **Interior Designer and Visualizer**
BETA Architects Pvt Ltd. Penang, Malaysia.

- Contributed to the Kedah Digital Library project, recipient of the Malaysian Institute of Architects (PAM) Gold Award and Malaysian Institute of Interior Designers (MIID) REKA Gold Award in Adaptive Reuse.
- Supported the design and development of three major renovation projects from concept through execution, including spatial planning, material selection, detailing, and tender documentation.

2014-2016 **Interior Designer**
Archint Design & Build Pvt Ltd. Penang, Malaysia.

- Led client consultations and site assessments to understand project requirements, spatial needs, and design preferences, and to establish the overall design direction.
- Developed interior design concepts, space plans, color schemes, 3D visualizations, and comprehensive proposals, followed by tender documentation, project schedules, and material and finish specifications.

REFEREED JOURNAL PUBLICATIONS

† *Equal contribution as first author; "IF" denotes Journal Impact Factor as of September 2026.*

[J12] **Ashour, M.**, Yang, Q., Qiu, Z., & Kalantari, S. (2026). Designing Human-Centered Journeys: Design Strategies for Enhancing Wayfinding in Complex Healthcare Environments. *HERD: Health Environments Research & Design Journal* (IF= 2.8). *19*(3), 9-30. https://doi:10.1177/19375867261421977.
Selected for EDAC and AIA Continuing Education Unit (CEU).

[J11] Shepley, M. M., Joyce, K., **Ashour, M.**, Radmanesh, N., Gao, Y., Liu, Z., ... & Thai, J. (2026). Staff, Patient, and Designer Perspectives on Outpatient Rehabilitation Therapy Environments: A Pilot Study. *HERD: Health Environments Research & Design Journal* (IF= 2.8). *19*(3), 184-207. https://doi:10.1177/19375867241304029.

[J10] Qiu, Z., **Ashour, M.**, Zhou, X., & Kalantari, S. (2024). NavMarkAR: A Landmark-Based Augmented Reality (AR) Wayfinding System for Enhancing Older Adults' Spatial Learning. *Advanced Engineering Informatics* (IF= 11.5). *62*, 102635. https://doi:10.1016/j.aei.2024.102635.

[J9] Wang, M., **Ashour, M.**, Mahdiyar, A., Sabri, S. (2024). Opportunities and Threats of Adopting Digital Twin in Construction Projects: A Review. *Buildings* (IF= 3.6). *14*(8), 2349. https://doi.org/10.3390/buildings14082349.

[J8] **Ashour, M.,** & Mahdiyar, A. (2024). A Comprehensive State-of-the-Art Survey on the Recent Modified and Hybrid Analytic Hierarchy Process Approaches. *Applied Soft Computing* (IF= 7.8). *150*, 111014. https://doi.org/10.1016/j.asoc.2023.111014.
Derived from master's thesis research.

[J7] Tabatabaee, S., **Ashour, M.,** Sadeghi, H., Hoseini, S. A., Mohandes, S. R., Mahdiyar, A., ... & Hosseini, M. R. (2024). Towards the Adoption of Most Suitable Green Walls Within Sustainable Buildings Using Interval Type-2 Fuzzy Best-Worst Method and TOPSIS Technique. *Engineering, Construction and Architectural Management* (IF= 4.5). *31*(3), 1145-1164. https://doi.org/10.1108/ECAM-06-2022-0551.

[J6] Motamedpooya, S.,† **Ashour, M.**,† Mahdiyar, A., & Jamei, E. (2023). Diagnosing the Cause-Effect Relationship Among Deterrents to Intensive Green Roof Adoption: A Hybrid EFDM-FDEMATEL Approach. *Sustainable Cities and Society* (IF= 13.3). *99*, 104972. https://doi.org/10.1016/j.scs.2023.104972.

[J5] Kabirifar, K., **Ashour, M.,** Yazdani, M., Mahdiyar, A., & Malekjafarian, M. (2023). Cybernetic-Parsimonious MCDM Modeling with Application to the Adoption of Circular Economy in Waste Management. *Applied Soft Computing* (IF= 7.8). *139*, 110186. https://doi.org/10.1016/j.asoc.2023.110186.

[J4] **Ashour, M.,** Mahdiyar, A., Haron, S. H., & Hanafi, M. H. (2022). Barriers to the Practice of Sustainable Interior Architecture and Design for Interior Renovations: A Parsimonious-Cybernetic Fuzzy AHP Approach. *Journal of cleaner production* (IF= 10.7). *366*, 132958. https://doi.org/10.1016/j.jclepro.2022.132958.
Derived from master's thesis research.

[J3] Durdyev, S., **Ashour, M.,** Connelly, S., & Mahdiyar, A. (2022). Barriers to the Implementation of Building Information Modelling (BIM) for Facility Management. *Journal of Building Engineering* (IF= 8.1). *46*, 103736. https://doi.org/10.1016/j.jobe.2021.103736.

[J2] Tabatabaee, S., **Ashour, M.**, Mohandes, SR., Sadeghi, H., Mahdiyar, A., Hosseini, M. R., & Ismail, S. (2022). Deterrents to the Adoption of Green Walls: A Hybrid Fuzzy-based Approach. *Engineering, Construction and Architectural Management* (IF= 4.5). *29* (9), 3460-3479. https://doi.org/10.1108/ECAM-04-2021-0286

[J1] **Ashour, M.,** Mahdiyar, A., & Haron, S. H. (2021). A Comprehensive Review of Deterrents to the Practice of Sustainable Interior Architecture and Design. *Sustainability* (IF= 4.1). *13*(18), 10403. https://doi.org/10.3390/su131810403. ***Derived from master's thesis research.***

Under Review:

[UR4] **Ashour, M.**, Yang, Q., Mostafavi, A., Snell, R., Wu, J., Gao, Y., Niu, W., & Kalantari, S. (under review). Effects of Window and Signage Presence on Wayfinding Route Choices in a Healthcare Setting. *Journal of Environmental Psychology* (IF= 7.3).

[UR3] Zhao, M., Contero, M., **Ashour, M.,** Gramann, K., & Kalantari, S. (under review). From Neurophysiological Signals to Design Preference: A Systematic Review of Measures, Constructs, and Validation Approaches in 3D Design Evaluation. *Design Studies* (IF= 4.6).

[UR2] **Ashour, M.**, Yang, Q., Qiu, Z., & Kalantari, S. (under review). Integrating Spatial Cognition Research, Environmental Psychology, and Expert Practice: Wayfinding by Design. *Brain and Environment* (IF= not available).

[UR1] **Ashour, M.**, Yang, Q., Qiu, Z., & Kalantari, S. (under review). Choreographing Wayfinding in the Healthcare Design Process: An Industry Informed Framework for Multidisciplinary Collaboration. *Architectural Engineering and Design Management* (IF= 2.8).

In Preparation:

[IP3] **Ashour, M.,** & Kalantari, S. (in preparation). Front-Loading Rigor in Reflexive Thematic Analysis: An Interactional Sensemaking Approach for Design Research with Expert Participants.

[IP2] **Ashour, M.,** Anderson, A., Wells, N. M., & Kalantari, S. (in preparation). Wayfinding as a Dynamic System: Integrating Cognition, Affect, and Environment.

[IP1] Xu, T., **Ashour, M.,** Yang, Q., & Kalantari, S. (in preparation). Real-World Healthcare Wayfinding Under Uncertainty: Affective Mechanisms and Behavioral Prediction.

REFEREED CONFERENCE PRESENTATIONS, POSTERS, AND WORKSHOPS

[CP6] **Ashour, M.,** Yang, Q., Qiu, Z., & Kalantari, S. (2026, May 27–30). Design Reasoning and Wayfinding Theory: Insights from Healthcare Practice [Presentation]. *Environmental Design Research Association 57th Annual Conference (EDRA57)*, Amherst, MA, United States.

[CP5] Mostafavi, A., **Ashour, M.,** & Kalantari, S. (2026, May 27–30). Embedded or Applied? Landmark Type Effects on Spatial Memory in a Simulated Hospital Environment [Presentation]. *Environmental Design Research Association 57th Annual Conference (EDRA57)*, Amherst, MA, United States.

[CP4] Mostafavi, A., **Ashour, M.,** & Kalantari, S. (2026, May 27–30). Game-Based Wayfinding as a Participatory Research Method for Healthcare Design [Workshop]. *Environmental Design Research Association 57th Annual Conference (EDRA57)*, Amherst, MA, United States.

[CP3] **Ashour, M.,** Yang, Q., Qiu, Z., & Kalantari, S. (2025, September 19). Wayfinding by Design: Theoretical Echoes in Expert Practice [Poster presentation]. *Seventh Biennial Academy of Neuroscience for Architecture (ANFA) Conference: Neuroscience for Architecture*, La Jolla, CA, United States.

[CP2] **Ashour, M.,** Yang, Q., Qiu, Z., & Kalantari, S. (2025, May 1). Designing Human-Centered Journeys: Strategies for Enhancing Wayfinding in Complex Healthcare Environments [Poster presentation]. *Bronfenbrenner Center for Translational Research Poster Symposium*, Ithaca, NY, United States.

[CP1] **Ashour, M.,** Mojtahedi, A. (2024, June 8). Human-Centered Design for Architecture: S, M, L, XL [Presentation]. *American Institute of Architects (AIA) Conference on Architecture & Design*, Washington, D.C., United States.

Under Review:

[UR2] **Ashour, M.**, Mostafavi, A., Wu, J., & Kalantari, S. (under review). Real Enough for What? A Systematic Review of Human Response Correspondence Between Immersive Virtual and Real Environments. *CHI Conference on Human Factors in Computing Systems (CHI '27)*.

[UR1] **Ashour, M.**, Mostafavi, A., Bahrami Dahi, A., Xu, T. B., Boot, W. R., Czaja, S. J., & Kalantari, S. (under review). How Task Structure Shapes Dyadic Social Interaction in Multiplayer VR for Older Adults with Differing Cognitive Abilities. *CHI Conference on Human Factors in Computing Systems (CHI '27)*.

GRANTS AND FELLOWSHIPS

2025 **Academy of Neuroscience for Architecture (ANFA), The Hay Research Grant Program**
$50,000 | **Honorable Mention** (not awarded)
Investigating the Cognitive and Affective Mechanisms of Nature-Integrated Wayfinding in Healthcare Environments

2026 **Cornell Center for Social Sciences (CCSS), Seed Grant, Co-PI** (under review)
$12,000
Exploring Links Between Childhood Socioeconomic Status and Spatial Competencies.

HONORS AND AWARDS

2026 Graduate Fellowship for Fall 2026, **Cornell University**, Ithaca, USA.

2024-2026 Three consecutive departmental summer Fellowship Awards, **Cornell University**, Ithaca, USA.

2023 Full Graduate Ph.D. Scholarship, **Cornell University**, Ithaca, USA.

2022 Best Thesis Award, master's thesis, **University of Science Malaysia**, Penang, Malaysia.

2021 Viva Voce Scale 1 (highest mark), master's thesis defense, **University of Science Malaysia**, Penang, Malaysia.

2019 2nd Place, Malaysian Institute of Interior Designers (MIID) Design Competition, **MIID**, Kuala Lumpur, Malaysia.

2018-2019 Two consecutive Best Design Student Awards, **University of Science Malaysia**, Penang, Malaysia.

2018 Nominee, Malaysian Institute of Interior Designers (MIID) Design Competition, **MIID**, Kuala Lumpur, Malaysia.

2016-2019 Five consecutive Dean's List Awards, **University of Science Malaysia**, Penang, Malaysia.

TEACHING EXPERIENCE

FA/SP/SU = Fall/Spring/Summer; Course codes: *DEA = Design and Environmental Analysis; IAED = Interior Architecture and Environmental Design.*

2026 **Studio Instructor of Record**
Department of Human Centered Design, **Cornell University,** Ithaca, USA.

- Design Generation(s) Studio (DEA 1100) | 18 students (SU-26)
  - Developed the syllabus, course structure, assignments, and assessment criteria for a three-week intensive design studio.
  - Designed and delivered seven lectures, workshops, and in-class exercises on design fundamentals, design philosophy, design thinking, and human-centered design.
  - Received strong student evaluations, including 4.53/5 for overall course quality, 4.67/5 for instructional support, and 4.87/5 for fostering a sense of belonging (83% response rate) all above department and school averages.

2023-pres. **Teaching Assistant**
Department of Human Centered Design, **Cornell University,** Ithaca, USA.

- **Studio Courses**
  - StudioSHIFT (DEA 2203) | 13 Students (SP-26)
  - Design Generation(s) Studio (DEA 1100) | 15, 17 Students (SU-24 and SU-25 | online)
  - Design Portfolio and Communication Studio (DEA 4050) | 17 Students
- **Contributions:**
  - Developed project briefs, assignment structures, and grading rubrics; independently led class sessions, lectures, and workshops (including AI-assisted concept exploration)
  - Mentored students through studio critiques, office hours, and written feedback

- **Lecture/seminar courses:**
  - Professional Practices and Ethics (DEA 4040) | 24 Students (SP-25)
  - Hospitality, Health, and Design Industry Immersion Seminar (DEA 6055) | 52 Students (SP-25)
  - Design Accountability: Evaluation of the Physical Environment (DEA 5304) | 6 Students (FA-23)
- **Contributions:**
  - Restructured seminar content and supported assessment and grading; led workshops on research design, data collection, and analytic planning
  - Coordinated faculty and industry engagement, portfolio reviews, and networking; coached a student research project to a co-authored peer-reviewed publication

2021-2023 Department of Interior Architecture and Environmental Design, **Bilkent University,** Ankara, Turkiye.

- **Studio Courses**
  - Interior Design Studio II (IAED 202) | 21 Students (SP-22)
  - Interior Design Studio I (IAED 201) | 21 Students (FA-21)
  - Sustainable Design for Interiors (IAED 342) | 18 Students (SP-21)
- **Contributions:**
  - Managed grading, exam proctoring, and course administration (including exam materials)
  - Provided individualized critiques during studio and office hours

INSTRUCTIONAL CONTRIBUTIONS

[GL] = Guest Lecture; [DR] = Design Review/Jury; [W] = Workshop; [P] = Presentation.

2026 [GL] *"Pre-Occupancy Evaluation: The Case of Wayfinding in Healthcare Settings"* in Health Facilities Planning Fundamentals for Managers/Entrepreneurs/Pre-Clinicians (PUBPOL 5740), **Cornell University,** Ithaca, USA.

2026 [GL] ***"****Designing Human-Centered Journeys: Cognitive Aging and the Making of Legible Environments*" in Designing Age Friendly Environments (DEA 5700), **Cornell University,** Ithaca, USA.

2026 [GL] ***"****Let's Talk Qualitative: Before codes, after themes, and everything in between"* in Studies in Design Thinking (DEA 6100), **Cornell University,** Ithaca, USA.

2024-2026 [DR] Mystery Day referee (twice) for Research Methods in Social Sciences (DEA 6560), **Cornell University,** Ithaca, USA.

2024 [P] ***"****Technology and Human Flourishing: From Traditional Foundations to Digital Horizons,"* Cornell Human Ecology Centennial 100 Film Series, **Cornell University,** Ithaca, USA.

2024 [DR] Schematic Design Review for Adaptive Reuse Studio (DEA 4401), **Cornell University,** Ithaca, USA.

2024 [GL] ***"****The Philosophy of Design"* in Design Portfolio and Communication (DEA 4050), **Cornell University,** Ithaca, USA.

2023 [DR] Mid-term Jury for Adaptive Reuse Studio (DEA 4401), **Cornell University,** Ithaca, USA.

2022, 2023 [DR] Final Jury for Interior Design Studio I and II (IAED 201, 202), **Bilkent University,** Ankara, Turkiye.

2022 [DR] Final Jury for Sustainable Design for Interiors (IAED 342), **Bilkent University,** Ankara, Turkiye.

2021 [W] ***"****Conducting Systematic Literature Reviews"* for graduate students in Construction Management program, **University of Science Malaysia**, Penang, Malaysia.

2021 [GL] ***"****Visualization: Applications in the Construction Industry"* in Information and Communication Technology in Project Management (RMG 561), **University of Science Malaysia**, Penang, Malaysia.

2020 [GL] ***"****3D Rendering, Postproduction, and Portfolio Design"* for students in Interior Design Program. **Georgia Southern University**, Statesboro, USA.

2018 [W] *"Visualization and Post-Production"* for designers at **Mieux Design Studio**, Kuala Lumpur, Malaysia.

2015 [T] *"Basic Introduction to Sketchup"* in Enspark Art & Design Talk, **UOW Malaysia KDU Penang University College**, Penang, Malaysia.

ACADEMIC AND DEPARTMENTAL SERVICES

2021-pres. **Ad Hoc Journal Reviewer:**

- Journal of Cleaner Production
- Frontiers in Built Environment
- CoDesign
- Discover Sustainability

- Health Environments Research & Design Journal
- JMIR Aging
- Sustainability
- Intelligent Buildings International

2025 **Departmental Academic Service Assistant for PURSUIT Career Fair**
Department of Human Centered Design, **Cornell University**, Ithaca, USA.
- Coordinated employer outreach, communications, branding, Portfolio Review Night, and event logistics.
- Engaged 81 firms; supported participation of 36 students who secured 59 interviews.
- Built a comprehensive employer database and structured filing system to strengthen employer engagement, institutional continuity, and future event planning.

2019 **Departmental Assistant for MIID Design Competition**
School of Housing, Building and Planning, **University of Science Malaysia**, Penang, Malaysia.
- Designed and fabricated wooden trophies for competition winners using laser cutting, and prepared plaques of appreciation, certificates, and invitations for competition jury and guests.

MEMBERSHIPS

Since 2025 Environmental Design Research Association (EDRA)

Since 2025 Academy of Neuroscience for Architecture (ANFA)

Since 2024 Cornell Center for Social Sciences (CCSS)